pandas

In [ ]:
```
Series    -> one dim data  (same datatype)
DataFrame -> two dim data / table like structure in row-col format
```

Series

In [3]:
```
pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\mania\anaconda3\lib\site-packag
es (2.3.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\mania\anaconda3\lib\site
-packages (from pandas) (2.2.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\mania\anaconda3
\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\mania\anaconda3\lib\site-
packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\mania\anaconda3\lib\sit
e-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\mania\anaconda3\lib\site-pack
ages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

In [5]:
```
import numpy as np
import pandas as pd
```

In [41]:
```
s1 = pd.Series(data = [10,20,30], index = range(101,104))
```

In [43]:
```
s1
```

Out[43]:
```
101    10
102    20
103    30
dtype: int64
```

In [47]:
```
s1[102] = 2000
```

In [49]:
```
s1
```

Out[49]:
```
101      10
102    2000
103      30
dtype: int64
```

In [25]:
```
print(type(s1))
```

```
<class 'pandas.core.series.Series'>
```

In [27]:
```
dict1 = {"a":10, "b":20, "c":30}
```

In [29]:
```
pd.Series(dict1)
```

Out[29]:
```
a    10
b    20
c    30
dtype: int64
```

In [31]:
```
arr = np.random.rand(35)
```

In [33]:
```
print(arr)
```

```
[0.55132102 0.12331588 0.23002503 0.65772401 0.53891015 0.74937164
 0.38320051 0.15315816 0.39570564 0.18069505 0.60557021 0.44111141
 0.32464734 0.91010328 0.3594683  0.10901659 0.36608644 0.93949617
 0.50522445 0.20224441 0.01675434 0.20192768 0.14667432 0.60802484
 0.82253952 0.87360535 0.74043348 0.57236172 0.08236978 0.70739391
 0.67198466 0.95228735 0.30530317 0.03901717 0.63508498]
```

In [35]:
```
pd.Series(arr)
```

Out[35]:
```
0     0.551321
1     0.123316
2     0.230025
3     0.657724
4     0.538910
5     0.749372
6     0.383201
7     0.153158
8     0.395706
9     0.180695
10    0.605570
11    0.441111
12    0.324647
13    0.910103
14    0.359468
15    0.109017
16    0.366086
17    0.939496
18    0.505224
19    0.202244
20    0.016754
21    0.201928
22    0.146674
23    0.608025
24    0.822540
25    0.873605
26    0.740433
27    0.572362
28    0.082370
29    0.707394
30    0.671985
31    0.952287
32    0.305303
33    0.039017
34    0.635085
dtype: float64
```

In [39]:
```
pd.Series(10,index = range(15))
```

Out[39]:
```
0     10
1     10
2     10
3     10
4     10
5     10
6     10
7     10
8     10
9     10
10    10
11    10
12    10
13    10
14    10
dtype: int64
```

Dataframe

In [52]:
```
d1 = {"Name":["Jay","Riya","Sanket"],
     "Marks" : [90,85,75],
     "City" : ["Surat","Rajkot","Rajkot"]}
```

In [54]:
```
d1
```

Out[54]:
```
{'Name': ['Jay', 'Riya', 'Sanket'],
 'Marks': [90, 85, 75],
 'City': ['Surat', 'Rajkot', 'Rajkot']}
```

In [64]:
```
df1 = pd.DataFrame(data = d1, index = range(101,104))
```

In [66]:
```
df1
```

Out[66]:

	Name	Marks	City
101	Jay	90	Surat
102	Riya	85	Rajkot
103	Sanket	75	Rajkot

In [68]:
```
print(type(df1))
```

```
<class 'pandas.core.frame.DataFrame'>
```

In [70]:
```
df1.describe()
```

Out[70]:

	Marks
count	3.000000
mean	83.333333
std	7.637626
min	75.000000
25%	80.000000
50%	85.000000
75%	87.500000
max	90.000000

In [72]:
```
df1.dtypes
```

Out[72]:
```
Name      object
Marks      int64
City      object
dtype: object
```

In [78]:
```
mr = np.random.randint(25,31,250).reshape(50,5)
```

In [90]:
```
df2 = pd.DataFrame(mr, index = range(101,151), columns = ["PDS","DM","IAT","CA",
```

In [96]:
```
df2.head(2)
```

Out[96]:

	PDS	DM	IAT	CA	Elective
101	30	28	29	30	27
102	28	29	30	28	25

In [100…
```
df2.tail(2)
```

Out[100…

	PDS	DM	IAT	CA	Elective
149	27	26	28	29	28
150	29	27	27	27	27

In [104…
```
df2.describe()
```

Out[104...

	PDS	DM	IAT	CA	Elective
count	50.000000	50.000000	50.000000	50.000000	50.000000
mean	27.540000	27.720000	27.220000	27.860000	27.580000
std	1.842913	1.591303	1.729575	1.714405	1.679286
min	25.000000	25.000000	25.000000	25.000000	25.000000
25%	26.000000	26.250000	26.000000	26.250000	26.000000
50%	28.000000	28.000000	27.000000	28.000000	27.500000
75%	29.000000	29.000000	29.000000	29.000000	29.000000
max	30.000000	30.000000	30.000000	30.000000	30.000000

In [106...

```
df2.head()
```

Out[106...

	PDS	DM	IAT	CA	Elective
101	30	28	29	30	27
102	28	29	30	28	25
103	25	29	25	30	26
104	27	28	27	30	26
105	27	27	30	27	27

In [116...

```
df2["PDS"][102] = 0
```

In [118...

```
df2["PDS"]
```

Out[118…

```
101    30
102     0
103    25
104    27
105    27
106    28
107    25
108    30
109    27
110    29
111    29
112    28
113    28
114    28
115    26
116    30
117    26
118    25
119    25
120    30
121    26
122    30
123    28
124    30
125    25
126    29
127    25
128    25
129    26
130    30
131    30
132    26
133    27
134    30
135    30
136    26
137    28
138    29
139    26
140    29
141    28
142    29
143    26
144    26
145    30
146    26
147    25
148    25
149    27
150    29
Name: PDS, dtype: int32
```

In [120…

```
df2.head()
```

Out[120…

	PDS	DM	IAT	CA	Elective
101	30	28	29	30	27
102	0	29	30	28	25
103	25	29	25	30	26
104	27	28	27	30	26
105	27	27	30	27	27

In [124…
```
pd.DataFrame(0,index = [1,2,3], columns = ["C1","C2"])
```

Out[124…

	C1	C2
1	0	0
2	0	0
3	0	0

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
    PDS IAT CN/CA DM
101
102
.
.
.
150
```

In [11]:
```
df = pd.DataFrame(data = np.random.randint(45,101,200).reshape(50,4), index = ra
```

In [17]:
```
df.head(3)
```

Out[17]:

	PDS	IAT	CA/CN	DM
101	68	99	85	50
102	81	94	100	96
103	54	99	97	94

In [19]:
```
df.tail()
```

Out[19]:

	PDS	IAT	CA/CN	DM
146	85	56	56	96
147	58	83	77	48
148	83	80	64	70
149	79	84	59	75
150	53	60	100	85

In [23]:
```
s = df['PDS']
```

In [25]:
```
print(type(s))
```

```
<class 'pandas.core.series.Series'>
```

In [35]:
```
df[['PDS','IAT']]
```

Out[35]:

	PDS	IAT
101	68	99
102	81	94
103	54	99
104	86	80
105	92	96
106	53	96
107	80	45
108	54	92
109	89	91
110	91	64
111	77	81
112	46	93
113	47	88
114	45	48
115	82	61
116	95	55
117	59	79
118	96	88
119	73	75
120	48	100
121	79	56
122	64	50
123	93	88
124	56	87
125	64	49
126	93	62
127	95	82
128	89	77
129	86	88
130	79	74
131	48	87
132	88	52
133	53	77

	PDS	IAT
134	95	90
135	81	82
136	74	75
137	49	46
138	45	70
139	53	88
140	53	52
141	62	94
142	59	75
143	53	63
144	76	58
145	86	65
146	85	56
147	58	83
148	83	80
149	79	84
150	53	60

In [39]:
```
df['PDS'][105] = 0
```

In [41]:
```
df.head()
```

Out[41]:

	PDS	IAT	CA/CN	DM
101	68	99	85	50
102	81	94	100	96
103	54	99	97	94
104	86	80	82	53
105	0	96	60	73

In [45]:
```
df.loc[103]
```

Out[45]:
```
PDS      54
IAT      99
CA/CN    97
DM       94
Name: 103, dtype: int32
```

In [53]:
```
df.loc[[125,150],['PDS','DM']]
```

Out[53]:

	PDS	DM
125	64	85
150	53	85

In [55]:
```
df[['DM']]
```

Out[55]:

	DM
101	50
102	96
103	94
104	53
105	73
106	64
107	47
108	82
109	85
110	53
111	68
112	76
113	88
114	45
115	45
116	96
117	93
118	68
119	54
120	48
121	81
122	51
123	71
124	78
125	85
126	91
127	94
128	65
129	57
130	66
131	77
132	81
133	85

	DM
134	81
135	73
136	65
137	81
138	62
139	56
140	72
141	93
142	49
143	93
144	81
145	58
146	96
147	48
148	70
149	75
150	85

In [57]:
```
df.loc[:,['DM']]
```

Out[57]:

	DM
101	50
102	96
103	94
104	53
105	73
106	64
107	47
108	82
109	85
110	53
111	68
112	76
113	88
114	45
115	45
116	96
117	93
118	68
119	54
120	48
121	81
122	51
123	71
124	78
125	85
126	91
127	94
128	65
129	57
130	66
131	77
132	81
133	85

	DM
134	81
135	73
136	65
137	81
138	62
139	56
140	72
141	93
142	49
143	93
144	81
145	58
146	96
147	48
148	70
149	75
150	85

In [59]:
```
df.head()
```

Out[59]:

	PDS	IAT	CA/CN	DM
101	68	99	85	50
102	81	94	100	96
103	54	99	97	94
104	86	80	82	53
105	0	96	60	73

In [71]:
```
df.iloc[[0,49],[0,2]]
```

Out[71]:

	PDS	CA/CN
101	68	85
150	53	100

In [79]:
```
df.loc[101,'ML'] = 10
```

In [81]:
```
df.head()
```

Out[81]:

	PDS	IAT	CA/CN	DM	ML
101	68	99	85	50	10.0
102	81	94	100	96	NaN
103	54	99	97	94	NaN
104	86	80	82	53	NaN
105	0	96	60	73	NaN

In [89]:
```
df.drop(101, inplace = True)
```

In [91]:
```
df.head()
```

Out[91]:

	PDS	IAT	CA/CN	DM	ML
102	81	94	100	96	NaN
103	54	99	97	94	NaN
104	86	80	82	53	NaN
105	0	96	60	73	NaN
106	53	96	51	64	NaN

In [97]:
```
df = df.drop('ML', axis = 1)
```

In [99]:
```
df.head()
```

Out[99]:

	PDS	IAT	CA/CN	DM
102	81	94	100	96
103	54	99	97	94
104	86	80	82	53
105	0	96	60	73
106	53	96	51	64

In [101…
```
df.index
```

Out[101…
```
RangeIndex(start=102, stop=151, step=1)
```

In [105…
```
df.columns
```

Out[105…
```
Index(['PDS', 'IAT', 'CA/CN', 'DM'], dtype='object')
```

In [107…
```
df.head()
```

Out[107…

	PDS	IAT	CA/CN	DM
102	81	94	100	96
103	54	99	97	94
104	86	80	82	53
105	0	96	60	73
106	53	96	51	64

In [113…
```
df.sort_index(ascending=False, inplace = True)
```

In [115…
```
df.head()
```

Out[115…

	PDS	IAT	CA/CN	DM
150	53	60	100	85
149	79	84	59	75
148	83	80	64	70
147	58	83	77	48
146	85	56	56	96

In [117…
```
df.sort_index(axis=1,inplace = True, ascending=False)
```

In [119…
```
df.head()
```

Out[119…

	PDS	IAT	DM	CA/CN
150	53	60	85	100
149	79	84	75	59
148	83	80	70	64
147	58	83	48	77
146	85	56	96	56

In [123…
```
arr = df.to_numpy()
```

In [127…
```
print(type(arr))
```

```
<class 'numpy.ndarray'>
```

In [129…
```
a1 = np.array(np.random.randint(10,25,10))
```

In [131…
```
a1
```

Out[131…
```
array([18, 18, 23, 13, 24, 13, 16, 15, 19, 21], dtype=int32)
```

In [135…
```
a1[(a1>20) & (a1%2==0)]
```

Out[135… array([24], dtype=int32)

In [137…
```
df[df>90]
```

Out[137...

	PDS	IAT	DM	CA/CN
150	NaN	NaN	NaN	100.0
149	NaN	NaN	NaN	NaN
148	NaN	NaN	NaN	NaN
147	NaN	NaN	NaN	NaN
146	NaN	NaN	96.0	NaN
145	NaN	NaN	NaN	94.0
144	NaN	NaN	NaN	93.0
143	NaN	NaN	93.0	96.0
142	NaN	NaN	NaN	91.0
141	NaN	94.0	93.0	NaN
140	NaN	NaN	NaN	NaN
139	NaN	NaN	NaN	NaN
138	NaN	NaN	NaN	NaN
137	NaN	NaN	NaN	NaN
136	NaN	NaN	NaN	NaN
135	NaN	NaN	NaN	NaN
134	95.0	NaN	NaN	NaN
133	NaN	NaN	NaN	NaN
132	NaN	NaN	NaN	92.0
131	NaN	NaN	NaN	NaN
130	NaN	NaN	NaN	NaN
129	NaN	NaN	NaN	NaN
128	NaN	NaN	NaN	NaN
127	95.0	NaN	94.0	93.0
126	93.0	NaN	91.0	100.0
125	NaN	NaN	NaN	96.0
124	NaN	NaN	NaN	NaN
123	93.0	NaN	NaN	NaN
122	NaN	NaN	NaN	NaN
121	NaN	NaN	NaN	NaN
120	NaN	100.0	NaN	93.0
119	NaN	NaN	NaN	NaN
118	96.0	NaN	NaN	NaN

	PDS	IAT	DM	CA/CN
117	NaN	NaN	93.0	NaN
116	95.0	NaN	96.0	NaN
115	NaN	NaN	NaN	NaN
114	NaN	NaN	NaN	NaN
113	NaN	NaN	NaN	NaN
112	NaN	93.0	NaN	NaN
111	NaN	NaN	NaN	NaN
110	91.0	NaN	NaN	NaN
109	NaN	91.0	NaN	97.0
108	NaN	92.0	NaN	NaN
107	NaN	NaN	NaN	NaN
106	NaN	96.0	NaN	NaN
105	NaN	96.0	NaN	NaN
104	NaN	NaN	NaN	NaN
103	NaN	99.0	94.0	97.0
102	NaN	94.0	96.0	100.0

In [143…

```
df[(df['PDS'] > 90) | (df['DM']>80)][['PDS','DM']]
```

Out[143…

	PDS	DM
150	53	85
146	85	96
144	76	81
143	53	93
141	62	93
137	49	81
134	95	81
133	53	85
132	88	81
127	95	94
126	93	91
125	64	85
123	93	71
121	79	81
118	96	68
117	59	93
116	95	96
113	47	88
110	91	53
109	89	85
108	54	82
103	54	94
102	81	96

In [145…

```
df.head()
```

Out[145…

	PDS	IAT	DM	CA/CN
150	53	60	85	100
149	79	84	75	59
148	83	80	70	64
147	58	83	48	77
146	85	56	96	56

In [147…

```
df.loc[151] = [0,0,0,0]
```

In [151…
```
df.tail()
```

Out[151…

	PDS	IAT	DM	CA/CN
105	0	96	73	60
104	86	80	53	82
103	54	99	94	97
102	81	94	96	100
151	0	0	0	0

In [1]:
```
import numpy as np
import pandas as pd
```

In [9]:
```
d1 = "24th August, 2026 7:30PM"
```

In [13]:
```
d = pd.to_datetime(d1)
```

In [ ]:
```
# 24-08-2026 85%
# 25-08-2026 90
# 26-08-2026 87
# 27-08-2026 97
# 28-08-2026 100
```

In [24]:
```
x = d + pd.to_timedelta(np.arange(5), unit = "D")
```

In [28]:
```
pd.Series([85,90,87,97,100], index = x)
```

Out[28]:
```
2026-08-24 19:30:00     85
2026-08-25 19:30:00     90
2026-08-26 19:30:00     87
2026-08-27 19:30:00     97
2026-08-28 19:30:00    100
dtype: int64
```

In [ ]:
```
3-8-2026 8:00am     27
5-8-2026 8:00am     25
7-8-2026 8:00am     26
```

In [38]:
```
d1 = "3rd August, 2026 8:00am"
```

In [40]:
```
d = pd.to_datetime(d1)
```

In [42]:
```
x = d + pd.to_timedelta(np.arange(0,5,2),unit = 'D')
```

In [44]:
```
x
```

Out[44]:
```
DatetimeIndex(['2026-08-03 08:00:00', '2026-08-05 08:00:00',
               '2026-08-07 08:00:00'],
              dtype='datetime64[ns]', freq=None)
```

In [46]:
```
pd.Series(index = x, data = [27,25,26])
```

Out[46]:
```
2026-08-03 08:00:00    27
2026-08-05 08:00:00    25
2026-08-07 08:00:00    26
dtype: int64
```

In [ ]:
```
Possible Values of Unit parameter of to_timedelta():

'D' -> Days
'h' -> Hours
'min' -> Minutes
's' -> Seconds
'ms' -> Miliseconds
'us' -> microseconds
'ns' -> nanoseconds
```

In [52]:
```
df = pd.read_excel("C:\\Users\\mania\\OneDrive\\Desktop\\nptel course.xlsx")
```

In [54]:
```
df.head()
```

Out[54]:

	Course ID	Discipline	Course Name	SME Name	Institute	Coordinating Institute	Durati
0	noc26-cs104	Computer Science and Engineering	Generative AI for Computer Vision	Prof. Arijit Sur	IIT Guwahati	IIT Guwahati	Wee
1	noc26-cs119	Computer Science and Engineering	Introduction to Machine Learning - IITKGP	Prof. Sudeshna Sarkar	IIT Kharagpur	IIT Kharagpur	8 Wee
2	noc26-cs133	Computer Science and Engineering	Introduction to Machine Learning	Prof. Balaraman Ravindran	IIT Madras	IIT Madras	Wee
3	noc26-cs138	Computer Science and Engineering	Deep Learning - IIT Ropar	Prof. Sudarshan Iyengar	IIT Ropar	IIT Madras	Wee
4	noc26-cs171	Computer Science and Engineering	Introduction to Large Language Models (LLMs)	Prof. Tanmoy Chakraborty,\nProf. Soumen Chakra...	IIT Delhi\nIIT Bombay	IIT Delhi	Wee

In [62]:
```
df[df["Duration"] == "12 Weeks"]
```

Out[62]:

	Course ID	Discipline	Course Name	SME Name	Institute	Coordinating Institute
0	noc26-cs104	Computer Science and Engineering	Generative AI for Computer Vision	Prof. Arijit Sur	IIT Guwahati	IIT Guwahati
2	noc26-cs133	Computer Science and Engineering	Introduction to Machine Learning	Prof. Balaraman Ravindran	IIT Madras	IIT Madras
3	noc26-cs138	Computer Science and Engineering	Deep Learning - IIT Ropar	Prof. Sudarshan Iyengar	IIT Ropar	IIT Madras
4	noc26-cs171	Computer Science and Engineering	Introduction to Large Language Models (LLMs)	Prof. Tanmoy Chakraborty,\nProf. Soumen Chakra...	IIT Delhi\nIIT Bombay	IIT Delhi
5	noc26-cs180	Computer Science and Engineering	Natural Language Processing	Prof. Pawan Goyal	IIT Kharagpur	IIT Kharagpur
6	noc26-cs181	Computer Science and Engineering	Deep Learning for Natural Language Processing	Prof. Pawan Goyal	IIT Kharagpur	IIT Kharagpur
7	noc26-cs95	Computer Science	Fundamentals of Generative AI and Large Langua...	Prof. Sriram Ganapathy,\nProf. Ashwini Kodipal...	IISc Bangalore\nPES University\nRV University	IISc Bangalore
9	noc26-cs102	Computer Science and Engineering	Foundations of Machine Learning	IIT Hyderabad	IIT Madras	12 Weeks
12	noc26-cs129	Computer Science and Engineering	Artificial Intelligence: Concepts and Techniques	NaN	NaN	NaN
13	noc26-cs132	Computer Science and Engineering	Artificial Intelligence: Search Methods for Pr...	NaN	NaN	NaN
14	noc26-cs174	Computer Science and Engineering	Deep Learning for Computer Vision	NaN	NaN	NaN
15	noc26-cs179	Computer Science and Engineering	Applied Accelerated Artificial Intelligence	NaN	NaN	NaN

	Course ID	Discipline	Course Name	SME Name	Institute	Coordinating Institute
16	noc26-cs182	Computer Science and Engineering	Artificial Intelligence: Foundations and Algor...	NaN	NaN	NaN
17	noc26-cs97	Computer Science and Engineering	Mathematical Foundations of Generative AI	NaN	NaN	NaN

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16          NaT
17          NaT
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